Mathematical notations usually express the idea more accurately and elegantly than words. In the study of line integrals, we need to integrate vector fields along a curve or a segment with specific endpoints. Thus, I invented the following notations:

- Segment from A to B: $[_A^B$, or even $[_{(a,b,c)}^{(x,y,z)}$ if we want to state the coordinates of the endpoints
- Arc from A to B: $(_A^B$, or even $(_{(a,b,c)}^{(x,y,z)}$.

These are very convenient. For example, it is easy to check that

$$\mathbf{F}(x,y) = (3x^2y^2 + 2xy^3, 2x^3y + 3x^2y^2)$$

is conservative. We know that a potential of a conservative vector field $\mathbf{F} = (M, N)$ is given by fixing A and integrating along any arc from A to P:

$$f(P) = \int_{(_A^P} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_A^P \mathbf{F} \cdot \mathrm{d}\mathbf{r}$$

or

$$f(x,y) = \int_{(a,b)}^{(x,y)} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_{(a,b)}^{(x,y)} M(x,y)\mathrm{d}x + N(x,y)\mathrm{d}y.$$

Thus, a potential of our $\mathbf{F}$ is given by integrating from $(0,0)$ to (x,y) along $[_{(0,0)}^{(x,0)}$ then $[_{(x,0)}^{(x,y)}$:

$$\begin{aligned} f(x,y) &= \int_{(0,0)}^{(x,y)} (3x^2y^2 + 2xy^3)\mathrm{d}x + (2x^3y + 3x^2y^2)\mathrm{d}y \\ &= \int_{[_{(0,0)}^{(x,0)}} (3x^2y^2 + 2xy^3)\mathrm{d}x + (2x^3y + 3x^2y^2)\mathrm{d}y \\ &\quad + \int_{[_{(x,0)}^{(x,y)}} (3x^2y^2 + 2xy^3)\mathrm{d}x + (2x^3y + 3x^2y^2)\mathrm{d}y \end{aligned}$$

$$
\begin{aligned}
&= \int_{[_{(0,0)}^{(x,0)}} (3x^2y^2 + 2xy^3)\mathrm{d}x + \int_{[_{(x,0)}^{(x,y)}} (2x^3y + 3x^2y^2)\mathrm{d}y \\
&= \int_0^x (3t^2 \cdot 0^2 + 2t \cdot 0^3)\mathrm{d}t + \int_0^y (2x^3t + 3x^2t^2)\mathrm{d}t \\
&= 0 + x^3y^2 + x^2y^3 = x^3y^2 + x^2y^3,
\end{aligned}
$$

where the second term of the first integral and the first term of the second integral after the second "=" are zero because along the segments $[_{(0,0)}^{(x,0)}$ and $[_{(x,0)}^{(x,y)}$ we have $\mathrm{d}y = 0$ and $\mathrm{d}x = 0$, respectively.

Of course this method applied to 3 dimensional vector fields

$$\mathbf{F}(x, y, z) = (M(x, y, z), N(x, y, z), H(x, y, z)).$$

Compare the above approach with the way Example 3 on §16.3 of the textbook is solved.